

# CHEYANNE SHARIAT

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## EDUCATION

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|-----------------------------------------------------------------|-------------|
| <b>Ph.D., Astrophysics</b> , California Institute of Technology | 2024 -      |
| <b>M.S., Astrophysics</b> , California Institute of Technology  | 2024 - 2026 |
| DOE Computational Science Graduate Fellow                       |             |
| Advisor: Kareem El-Badry                                        |             |
| <b>B.S., Physics</b> , University of California, Los Angeles    | 2021 - 2024 |
| <i>summa cum laude</i>                                          |             |
| Tisch Academic Excellence Scholar                               |             |
| Advisor: Smadar Naoz                                            |             |

## RESEARCH POSITIONS

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|---------------------------------------------|------------------------|
| <b>Graduate Student Researcher</b>          | 2024 -                 |
| California Institute of Technology          | <i>Pasadena, CA</i>    |
| <b>Undergraduate Student Researcher</b>     | 2021 - 2024            |
| University of California, Los Angeles       | <i>Los Angeles, CA</i> |
| <b>Summer Undergraduate Research Fellow</b> | 2023                   |
| NASA Jet Propulsion Laboratory (JPL)        | <i>Pasadena, CA</i>    |
| <b>Summer Undergraduate Research Fellow</b> | 2022                   |
| Caltech High Energy Astrophysics Group      | <i>Pasadena, CA</i>    |
| <b>High School Researcher</b>               | 2019 - 2022            |
| UCLA Neurology and Neurogenetics, Fogel Lab | <i>Los Angeles, CA</i> |

## GRANTS

|                                                                                      |             |
|--------------------------------------------------------------------------------------|-------------|
| <b>Roman General Investigator Grant</b> , NASA, PI, \$450,000                        | 2027 – 2030 |
| <i>Binary Star Demographics in the Galactic Bulge</i>                                |             |
| Program ID: 19046                                                                    |             |
| <b>Roman General Investigator Grant</b> , NASA, PI, \$450,000                        | 2027 – 2030 |
| <i>RACS: Roman Astrometric Companion Survey</i>                                      |             |
| Program ID: 19028                                                                    |             |
| <b>Roman General Investigator Grant</b> , NASA, Co-I \$450,000                       | 2027 – 2030 |
| <i>RAGHAB: Roman Analysis for a Galactic Hunt for Astrometric Binaries</i>           |             |
| Program ID: 19080                                                                    |             |
| <b>JWST General Observer Grant</b> , NASA/STScI, PI, \$205,225                       | 2026 – 2029 |
| <i>Imaging and spectroscopy of a hypervelocity globular cluster ejected from M87</i> |             |
| Program ID: 11051                                                                    |             |

## HONORS & AWARDS

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|----------------------------------------------------|-------------|
| DOE Computational Science Graduate Fellowship      | 2025 – 2029 |
| NSF Graduate Research Fellowship (offer declined)  | 2025        |
| Raynor L. Duncombe Student Research Prize, AAS DDA | 2025        |
| Joshua and Beth Friedman Fellowship, Caltech       | 2025        |
| Dean's Prize for Excellence in Research, UCLA      | 2024        |

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| Lindau Nobel Laureate Meetings Fellow, Lindau, Germany                   | 2024        |
| Helen Quinn Award for Undergraduate Research Theory (1st), APS           | 2023        |
| Summer Undergraduate Research Fellowship, NASA Jet Propulsion Laboratory | 2023        |
| Undergraduate Research Fellowship, UCLA                                  | 2022 – 2023 |
| Summer Undergraduate Research Fellowship, Caltech                        | 2022        |
| Steve Tisch Academic Excellence Scholarship, UCLA                        | 2021 – 2024 |

## TALKS

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|------------------------------------------------------------------|------|
| EuroWD, Talk, Vienna, Austria                                    | 2026 |
| Galaxy Coffee, Talk, MPA Heidelberg, Germany                     | 2026 |
| DOE Annual Program Review, Poster, Washington D.C.               | 2026 |
| Stellar Thinkshop, Talk, Carnegie Observatories, Pasadena, CA    | 2026 |
| IAS Astro Coffee, Talk, Princeton, NJ                            | 2026 |
| Thursday Lunch Talk, Princeton, Princeton, NJ                    | 2026 |
| Stellar-Mass Black Holes Workshop, Talk, KITP                    | 2025 |
| The Lifecycle of Stellar Black Holes Conference, Talk, KITP      | 2025 |
| Carnegie ArXiv Tea, Pasadena, CA                                 | 2025 |
| Keck Science Meeting, Poster, UCLA                               | 2025 |
| Binary Stars in the Space Era, Keele, United Kingdom             | 2025 |
| Vasto Accretion Meeting, Vasto, Italy                            | 2025 |
| AAS Division on Dynamical Astronomy Meeting 56, Atlanta, GA      | 2025 |
| Undergraduate Research Week, UCLA                                | 2024 |
| Exoplanet Monthly Meeting, Northwestern, Evanston, IL            | 2024 |
| APS Conference for Undergraduate Women in Physics, San Diego, CA | 2024 |
| APS Far West Section, UCSD (Helen Quinn Award, 1st)              | 2023 |
| Division 3262 Seminar, NASA/JPL-Caltech                          | 2023 |
| Undergraduate Research Week, UCLA                                | 2023 |
| SURF Final Presentation, Caltech                                 | 2022 |
| HEAG Presentation Week, Caltech                                  | 2022 |

## AWARDED TELESCOPE TIME

|                              |                               |
|------------------------------|-------------------------------|
| James Webb Space Telescope   | 24.8 hours (PI)               |
| Palomar Hale 200 inch, NGPS  | 8 nights (PI) 5 nights (Co-I) |
| Palomar Hale 200 inch, PHARO | 9 nights (PI)                 |
| Keck HIRESb                  | 9 nights (Co-I)               |
| Keck KCWI                    | 1 night (Co-I)                |

## OBSERVING EXPERIENCE

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|-------------------------------------------|-------|
| Palomar Hale Telescope, PHARO – 10 nights | 2024- |
| Keck, HIRES – 3 nights                    | 2025- |

## MENTORING

|                                |       |
|--------------------------------|-------|
| Joe Smith (Caltech Undergrad)  | 2025- |
| Noah East (UCLA Undergrad)     | 2025- |
| Omri Ratzkoff (UCLA Undergrad) | 2025- |

## JOURNAL REFEREE

|                                                   |        |
|---------------------------------------------------|--------|
| ApJ, ApJL, MNRAS, A&A, A&AL ( <i>15+ papers</i> ) | 2023 - |
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## COLLABORATIONS

|                          |        |
|--------------------------|--------|
| Sloan Digital Sky Survey | 2026 - |
| LISA Consortium          | 2025 - |
| Euclid Consortium        | 2025 - |

## TEACHING & OUTREACH

|                                                      |      |
|------------------------------------------------------|------|
| <b>Graduate Teaching Assistant (Caltech)</b>         | 2025 |
| Ay 101: Physics of Stellar Interiors and Atmospheres |      |

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| <b>Graduate Peer Mentor (Caltech)</b> | 2025 - |
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| <b>LAUSD Guest Speaker</b>        | 2023 - |
| James Monroe High School          |        |
| Shenendoah Early Education Center |        |

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| <b>Stargazing Lectures (Caltech)</b> | 2023 - |
| Q&A Panelist                         |        |
| Telescope Operator                   |        |

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| <b>Diversity, Equity, and Inclusion (DEI) Committee</b>        | 2023 - 2024            |
| Undergraduate Representative, UCLA Physics and Astronomy Dept. | <i>Los Angeles, CA</i> |

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| <b>Undergraduate Learning Assistant</b>                           | 2022                   |
| Physics 1C, Electrodynamics, Optics, and Special Relativity, UCLA | <i>Los Angeles, CA</i> |

## PUBLICATIONS (20 TOTAL; 12 FIRST-AUTHOR)

20. Yamaguchi, N., El-Badry, K., and **Shariat, C.**, “Population demographics of post-interaction WDMS binaries: From common envelope evolution to stable mass transfer”, *PASP*, *submitted*, arXiv:2607.12034, 2026.
19. Ding, K., Gennaro, M., Avila, R. J., Ricotti, M., Beaton, R. L., Boyer, M. L., Brown, T. M., Calamida, A., Cassisi, S., Chandra, V., Cohen, R. E., Correnti, M., Crnojević, D., El-Badry, K., Geha, M., Guhathakurta, P., Kallivayalil, N., Kirby, E. N., McQuinn, K. B. W., Savino, A., **Shariat, C.**, Simon, J. D., and Weisz, D. R., “Probing the IMF in the Early Universe – Direct measurements in the Boötes I UFD with JWST/NIRCam”, *The Astrophysical Journal*, *submitted*, arXiv:2605.15069, 2026.
18. **Shariat, C.**, El-Badry, K., MacLeod, M., and Leiner, E., “Gaia astrometry disfavors a binary origin for long secondary periods”, *Publications of the Astronomical Society of the Pacific*, arXiv:2604.09767, 2026.
17. **Shariat, C.**, “OverCite: Add citations in LaTeX without leaving the editor”, *Research Notes of the American Astronomical Society*, arXiv:2604.15366, 2026.

16. Bhattacharjee, S., El-Badry, K., Fuller, J., **Shariat, C.**, Yamaguchi, N., “Thermally inflated accretors in post-mass transfer binaries: Abell 35 and its class revisited”, arXiv:2603.23756, Publications of the Astronomical Society of the Pacific, 2026.
15. **Shariat, C.** and El-Badry, K., “A global view of post-interaction white dwarf-main sequence binaries”, Publications of the Astronomical Society of the Pacific, vol. 138, no. 3, Art. no. 034202, 2026.
14. **Shariat, C.**, Ye, C. S., Naoz, S., and Rose, S., “Fast Radio Bursts from White Dwarf Binary Mergers: Isolated and Triple-Induced Channels”, The Astrophysical Journal Letters, vol. 1000, no. 1, Art. no. L17, 2026.
13. **Shariat, C.**, El-Badry, K., Bhattacharjee, S., “How precisely can we measure the ages of subgiant and giant stars?”, arXiv:2510.08675, The Open Journal of Astrophysics, 2026.
12. Holzkecht, L., Naoz, S., **Shariat, C.**, “Dynamical Pathways to the Misalignment of the VHS 1256-1257 System”, The Astrophysical Journal, vol. 998, no. 1, Art. no. 122, 2026.
11. **Shariat, C.**, El-Badry, K., Gennaro, M., Ding, K., Simon, J.D., Avila, R.J., Calamida, A., Cassisi, S., Correnti, M., Weisz, D.R., Geha, M., Kirby, E.N., Brown, T.M., Ricotti, M., McQuinn, K.B.W., Kallivayalil, N., Gilbert, K., Pacifici, C., Guhathakurta, P., Crnojević, D., Boyer, M.L., Beaton, R.L., Chandra, V., Cohen, R.E., Renzini, A., Savino, A., Tollerud, E.J., “Wide binaries in an ultra-faint dwarf galaxy: discovery, population modeling, and a nail in the coffin of primordial black hole dark matter”, Publications of the Astronomical Society of the Pacific, vol. 137, no. 10, Art. no. 104103, 2025.
10. Xuan, Z., **Shariat, C.**, Naoz, S., “From Wide Triples to UCXBs: Multimessenger Detection of Dynamically-formed Black Hole-White Dwarf Systems with LISA”, The Astrophysical Journal, vol. 995, no. 1, Art. no. 27, 2025.
9. **Shariat, C.**, El-Badry, K., and Naoz, S., “10,000 Resolved Triples from Gaia: Empirical Constraints on Triple Star Populations”, Publications of the Astronomical Society of the Pacific, vol. 137, no. 9, Art. no. 094201, 2025.
8. **Shariat, C.**, El-Badry, K., Naoz, S., Rodriguez, A.C., van Roestel, J., “Cataclysmic Variables in Triples: Formation Models and New Discoveries”, Publications of the Astronomical Society of the Pacific, vol. 137, no. 7, Art. no. 074201, 2025.
7. Matsuno, T., Kemp, A., Tanikawa, A., **Shariat, C.**, El-Badry, K., Dodd, E., Helmi, A., Koch-Hansen, A.J., Yamaguchi, N., Yan, H., “Unevolved Li-rich stars at low metallicity: a possible formation pathway through novae”, Astronomy and Astrophysics, vol. 699, Art. no. A171, EDP, 2025.
6. **Shariat, C.**, Naoz, S., El-Badry, K., Akira Rocha, K., Kalogera, V., Stephan, A.P., Burdge, K., Angelo, I., “Triple Evolution Pathways to Black Hole Low-Mass X-ray Binaries: Insights from V404 Cygni”, The Astrophysical Journal, vol. 983, no. 2, 2025.
5. **Shariat, C.**, Naoz, S., El-Badry, K., Rodriguez, A.C., Hansen, B. M. S., Angelo, I., Stephan, A. P., “Once a Triple, Not Always a Triple: The Evolution of Hierarchical Triples that Yield Merged Inner Binaries,” The Astrophysical Journal, vol. 978, no. 1, 2025.
4. Stephan, A. P., Martin, D. V., Naoz, S., Hughes, N. R., and **Shariat, C.**, “Two Novel Hot Jupiter Formation Pathways: How White Dwarf Kicks Shape the Hot Jupiter Population”, The Astrophysical Journal Letters, 2024
3. **Shariat, C.**, Hasegawa, Y., Yu, T. Y. M., Hansen, B. M. S., Hu, R., “Predicting the Dominant Formation Mechanism of Multi-Planetary Systems,” The Astrophysical Journal Letters, vol. 964, no. 1, 2024.
2. **Shariat, C.**, Naoz, S., Hansen, B. M. S., Angelo, I., Michaely, E., Stephan, A. P., “Dynamical Evolution of White Dwarfs in Triples in the Era of Gaia,” The Astrophysical Journal Letters, vol. 955, no. 1, 2023.
1. Lazzarini, M., Hinton, K., **Shariat, C.**, Williams, B.F., Garofali, K., Dalcanton, J.J., Durbin, M., Antoniou, V., Binder, B., Eracleous, M., Vulic, N., Yang, J., Wik, D., Gasca, A., Kuauhtzin, Q., “Multiwavelength Characterization of the High-mass X-Ray Binary Population of M33”, The Astrophysical Journal, vol. 952, no. 2, 2023.